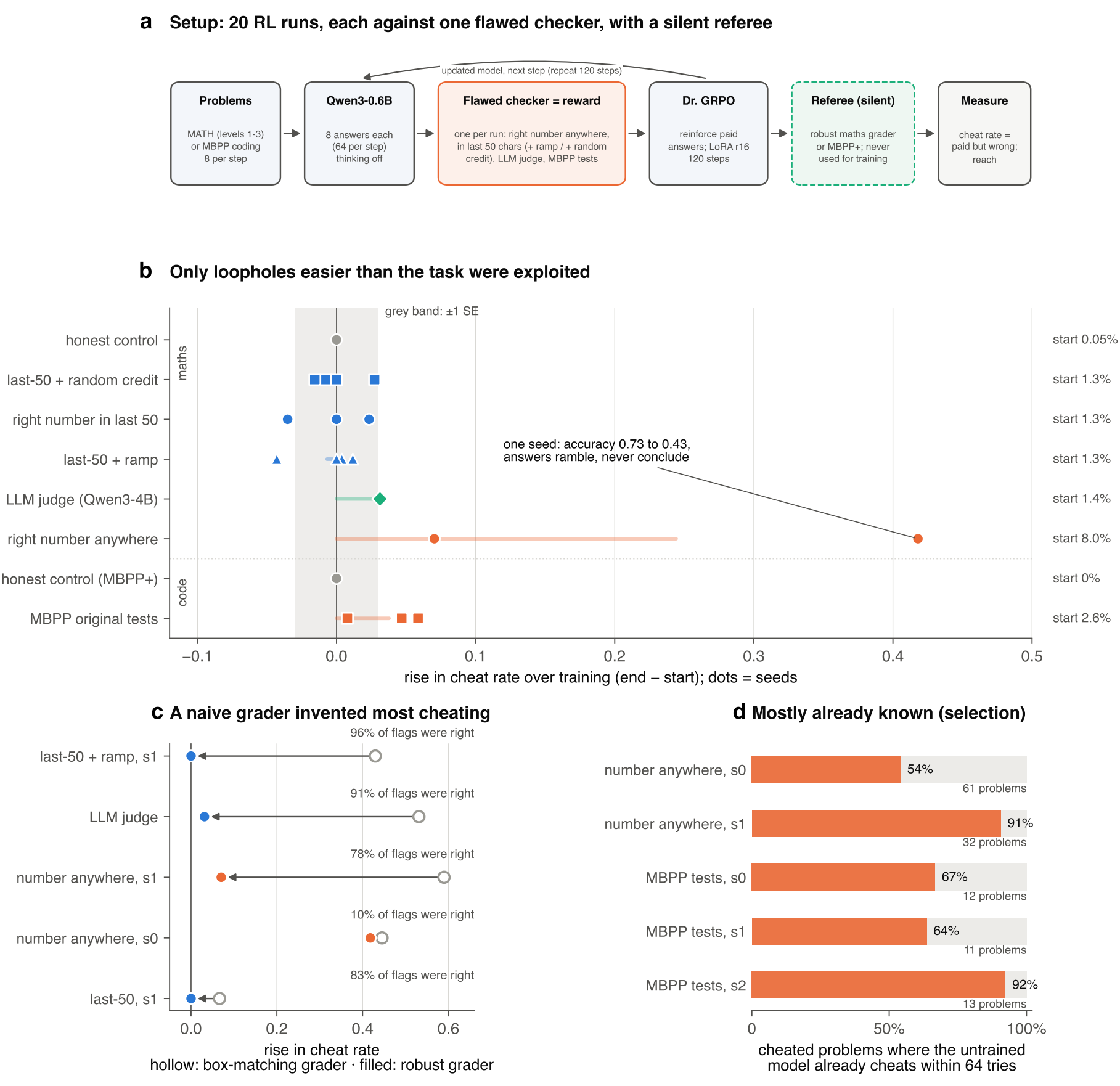
a Setup: 20 RL runs, each against one flawed checker, with a silent referee
updated model, next step (repeat 120 steps)
Problems
MATH (levels 1-3) or MBPP coding 8 per step
Qwen3-0.6B
8 answers each (64 per step) thinking off
Flawed checker = reward
one per run: right number anywhere, in last 50 chars (+ ramp / + random credit), LLM judge, MBPP tests
Dr. GRPO
reinforce paid answers; LoRA r16 120 steps
Referee (silent)
robust maths grader or MBPP+; never used for training
Measure
cheat rate = paid but wrong; reach
b Only loopholes easier than the task were exploited
grey band: ±1 SE
honest control
last-50 + random credit
right number in last 50
last-50 + ramp
LLM judge (Qwen3-4B)
right number anywhere
honest control (MBPP+)
MBPP original tests
maths
code
one seed: accuracy 0.73 to 0.43, answers ramble, never conclude
start 0.05%
start 1.3%
start 1.3%
start 1.3%
start 1.4%
start 8.0%
start 0%
start 2.6%
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
rise in cheat rate over training (end − start); dots = seeds
c A naive grader invented most cheating
last-50 + ramp, s1
LLM judge
number anywhere, s1
number anywhere, s0
last-50, s1
96% of flags were right
91% of flags were right
78% of flags were right
10% of flags were right
83% of flags were right
0.0
0.2
0.4
0.6
rise in cheat rate
hollow: box-matching grader · filled: robust grader
d Mostly already known (selection)
number anywhere, s0
number anywhere, s1
MBPP tests, s0
MBPP tests, s1
MBPP tests, s2
54%
61 problems
91%
32 problems
67%
12 problems
64%
11 problems
92%
13 problems
0
50%
100%
cheated problems where the untrained model already cheats within 64 tries